

MAN-MACHINE

The Brain

Brain computer interfaces such as a neural lace can directly connect computers to human brains. There are already interfaces that allow people to control robotic arms through a net of electrodes that reads electrical activity within the brain. The Halo Sport is a headpiece that improves learning and hand eye coordination by sending low intensity currents through the brain, a process known as neuro-priming.



The Spinal Cord

Over 14,000 people get spinal cord implants every year. These relatively simple devices consist of a remote, some wires, and a battery. Mild electrical impulses directly stimulate the spinal cord, and reduce the sensation of pain. The implants prevent the signals of pain originating in the body, from reaching the brain. New flexible implants in development could go a step further, and allow paralysed people to walk again, and restore bladder control.



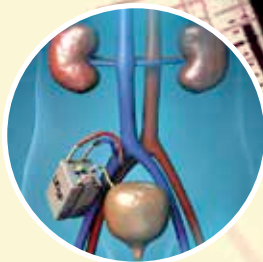
The Lungs

Replacing the lungs within the human body is too tricky to consider right now. However, medical devices that breathe like real lungs are needed considering the high incidence of respiratory diseases. Researchers from the University of Pittsburgh are developing an Extracorporeal membrane oxygenation (ECMO) device, that can work like a real lung from outside the body. It is designed to be carried around in a backpack.



The Kidneys

External dialysis machines already clean the blood for people with renal problems. The wait times for a kidney transplant are really long, and not everyone finds a suitable donor easily. Researchers from the University of California, San Francisco are working on creating an artificial kidney that can be integrated with the human body. One of the biggest challenges is ensuring that the blood does not clot while flowing through the artificial organ.



The Genes

Modifying human genes has many ethical concerns. In *Remaking Eden: Cloning and Beyond in a Brave New World*, Lee Silver outlines a vision of a dystopian future where genetically modified humans are the upper class, while the unmodified humans form a lower class. Premarital genetic screening tests already allow potential partners to check for risks of hereditary conditions, as well as receive genetic counselling.

